

Complete-cluster quotient, conditional covariance, and the nonvacuous Carleson frontier

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope

R-107 left a same-root adapted complete-cluster or equivalent direct source-action lower bound as the finite-cutoff, fixed-positive-floor frontier. The present package does not close that estimate. It performs the necessary quotient audit before another atom is proposed and gives the strongest exact normal form presently available.

The result has seven parts.

1. The historical R-085 Cartan estimate (4.10)–(4.11) is separated from the exact R-084 target. R-087 closed its regular spatial half, while R-088 removed an outer root weight and reduced the actual sufficient bridge to an unweighted one with any positive spatial gain. Neither historical nor direct sequential bridge is proved for progressive same-root feedback.
2. The historical rational form $F_{6.5}$ and the later fixed-chart coordinate K_R do not descend to the progressive subdivision quotient. They change sign on two representations of one exact production-fibre endpoint. The complete endpoint is invariant.
3. The quotient-safe complete rational endpoint has an exact conditional mean/covariance identity and an exact Cameron–Martin minimization. Its two surviving debts have no algebraic sign.
4. Every complete Hilbert-space cluster packet has an exact conditional mean/covariance normal form. This forces the mean square and signed covariance/trace defect to remain coupled.
5. The R-107 one-pair cluster proves that a covariance must be squared at the realized conditional-cluster level before the outer expectation. Averaging first misses the exact log-Laplace leading term. A parent sextic can repair the fixture only through an explicit quantified tradeoff.
6. An oscillatory adapted selector has bounded source energy and terminal sextic but an unbounded positive future-feedback tangent square. Thus no absolute future-feedback HS/PSD-Carleson extension can be paid from those two budgets alone. The complete signed second jet cancels, so this is not a counterexample to the signed action target.
7. The remaining direct and determinant-compatible targets are rewritten with the missing uniformity, sign, measurability, and order-of- operations requirements. The exact $q = 10/9$ budget is recorded.

The conclusion is a quotient-safe exact synthesis and method boundary, not a new lower-bound theorem. $\text{OVERLAP}_{\text{src}}$, Nelson, cutoff and floor removals, the interacting measure, and Sector A remain open. Tier stays T4.

2 The historical R-085 Cartan contract is not the current target

R-085 introduced the complete mixed production atom $u_{A,j,k,t}$ and asked for

$$\sum_A \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m u_{A,j,k,t}\|_{\mathfrak{S}_2}^2 dt \leq K 2^{-2s(m-k)} q_k, \quad s > \frac{1}{2}, \quad (2.1)$$

$$\sum_k 2^k q_k \lesssim 1 + \mathbb{E}[X^{1/2} Y^{1/2}]. \quad (2.2)$$

Its weighted Schur constant is

$$C_{s,\eta}^{\text{old}} = \frac{1}{(1-2^{-\eta})(1-2^{-2s})(1-2^{1+\eta-2s})}. \quad (2.3)$$

At $s = 7/12$, $\eta = 1/12$ this equals

$$C_{7/12,1/12}^{\text{old}} = 572.4472106721531 \dots \quad (2.4)$$

R-087 proves the spatial estimate (2.1) for the regular strict-past model. It does not prove (2.2): extracting a pathwise translated-model norm creates the registered rare-event growth. More importantly, R-088 audits the exact R-084 OU target and removes the outer 2^j weight which generated (2.2). The direct sufficient architecture is instead

$$\sum_A \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m v_{A,j,k,t}\|_{\mathfrak{S}_2}^2 dt \leq K 2^{-2s(m-k)} q_k, \quad s > 0, \quad (2.5)$$

$$\sum_k q_k \lesssim 1 + \mathbb{E}[X^{1/2} Y^{1/2}], \quad (2.6)$$

with

$$C_{s,\eta}^{\text{dir}} = \frac{1}{(1-2^{-\eta})(1-2^{-2s})(1-2^{\eta-2s})}. \quad (2.7)$$

At the balanced $s = \eta = 7/12$,

$$C_{7/12,7/12}^{\text{dir}} = 16.30295538482827 \dots \quad (2.8)$$

Multiplication by $q = 10/9$ gives respectively 636.0524563023923... and 18.11439487203141...

Equations (2.5)–(2.6) are a corrected sufficient target, not a theorem of this note. Their exact sequential production bridge remains open. R-105 further proves that a common-root source subdivision must be recombined to one coarse endpoint before a square is taken. Therefore a positive per-subvisit extension of either (2.2) or (2.6) is not legal.

3 Historical rational owners versus the subdivision quotient

Use the corrected R-105 notation

$$B_T = B_0 + DB_0[a] + \frac{1}{2} D^2 B_0[a, a], \quad L = B_1 - B_T, \quad (3.1)$$

$$Q = G^{\otimes 2} - \Gamma, \quad Q^a = (G + c)^{\otimes 2} - \Gamma. \quad (3.2)$$

The labelled fixed-chart owners are

$$\mathcal{R}_Q = \frac{1}{2} (B_1 - B_0) : Q, \quad \mathcal{M}_U = G^T B_T c, \quad (3.3)$$

$$\mathcal{K}_R = G^T L c + \frac{1}{2} c^T B_1 c. \quad (3.4)$$

Proposition 3.1 (the historical form is not the complete owner)

On one fixed chart,

$$\boxed{F_{6.5} = \frac{1}{2}L : Q^a + \frac{1}{2}c^T B_T c = \frac{1}{2}L : Q + \mathcal{K}_R.} \quad (3.5)$$

The quotient-safe complete endpoint is

$$\boxed{\Delta W_R = \frac{1}{2}B_1 : Q^a - \frac{1}{2}B_0 : Q = \mathcal{R}_Q + \mathcal{M}_U + \mathcal{K}_R.} \quad (3.6)$$

Moreover

$$\mathcal{R}_Q = \frac{1}{2} \left(DB_0[a] + \frac{1}{2}D^2 B_0[a, a] + L \right) : Q. \quad (3.7)$$

Thus $L : Q/2$ already belongs to the indivisible raw-Wick endpoint owner. Retaining $\mathcal{R}_Q$ and separately charging $F_{6.5}$ duplicates this term. Equation (3.5) remains a valid one-chart identity; it is superseded as a progressive owner.

Exact production-fibre subdivision fixture

Put

$$b(x) = \frac{4x^2(4x^2 + 9)^2}{81(1 + x^2)^2}. \quad (3.8)$$

Then

$$\begin{aligned} b(0) &= b'(0) = 0, \quad b''(0) = 8, \\ b(1) &= \frac{169}{81}, \quad b'(1) = \frac{208}{81}, \quad b''(1) = -\frac{2}{81}, \quad b(2) = \frac{400}{81}. \end{aligned} \quad (3.9)$$

Compare the single chart

$$(u, G) = (0, 1), \quad (a, c) = (2, 2), \quad \Gamma = 1,$$

with the two-step representation

$$(0, 1) \xrightarrow{(1,1)} (1, 2) \xrightarrow{(1,1)} (2, 3). \quad (3.10)$$

Direct substitution gives the following table, denoted (3.11):

	one chart	two-step sum	one minus sum
$\mathcal{R}_Q$	0	77/18	-77/18
$\mathcal{M}_U$	32	1076/81	1516/81
$\mathcal{K}_R$	-992/81	355/162	-2339/162
$F_{6.5}$	-992/81	427/162	-2411/162
ΔW_R	1600/81	1600/81	0.

The labelled defects cancel exactly:

$$-\frac{77}{18} + \frac{1516}{81} - \frac{2339}{162} = 0. \quad (3.12)$$

Hence both $F_{6.5}$ and $\mathcal{K}_R$ change sign under a representation-preserving subdivision, whereas ΔW_R is invariant. The production floor and covariance rescaling multiply the table by one positive factor and preserve these conclusions. This is a coordinate method no-go, not a complete torus-action counterexample.

4 Quotient-safe complete endpoint conditional identity

Let $\mathcal{H}$ contain the complete endpoint operators B_0, B_1 , the shift c , and the reference covariance Γ . Set

$$\mu = \mathbb{E}[G \mid \mathcal{H}], \quad V = \text{Cov}(G \mid \mathcal{H}). \quad (4.1)$$

All output contractions inside the cluster are retained in the block operators and in V .

Theorem 4.1 (complete endpoint conditional mean/covariance)

For the complete endpoint (3.6),

$$\boxed{\begin{aligned} \mathbb{E}[\Delta W_R \mid \mathcal{H}] &= \frac{1}{2}(c + \mu)^T B_1 (c + \mu) - \frac{1}{2}\mu^T B_0 \mu \\ &\quad + \frac{1}{2}(B_1 - B_0) : (V - \Gamma). \end{aligned}} \quad (4.2)$$

Proof. Conditional Pythagoras gives

$$\begin{aligned} \mathbb{E}[G^{\otimes 2} \mid \mathcal{H}] &= V + \mu^{\otimes 2}, \\ \mathbb{E}[(G + c)^{\otimes 2} \mid \mathcal{H}] &= V + (c + \mu)^{\otimes 2}. \end{aligned} \quad (4.3)$$

Insert (4.3) into (3.6). No owner is split. $\square$

Corollary 4.2 (exact Cameron–Martin minimization)

For $\eta > 0$,

$$\boxed{\begin{aligned} \inf_c \mathbb{E}[\Delta W_R + \eta \|c\|^2 \mid \mathcal{H}] &= \frac{1}{2}(B_1 - B_0) : (V - \Gamma) \\ &\quad + \frac{1}{2}\mu^T \left[2\eta B_1 (B_1 + 2\eta \text{I})^{-1} - B_0 \right] \mu. \end{aligned}} \quad (4.4)$$

Indeed the unique minimum on the active range is attained at

$$c_* = -(B_1 + 2\eta \text{I})^{-1} B_1 \mu. \quad (4.5)$$

At $q = 10/9$ and $\eta = 1/(2q) = 9/20$,

$$\boxed{\begin{aligned} \inf_c \left\{ q \mathbb{E}[\Delta W_R \mid \mathcal{H}] + \frac{1}{2} \|c\|^2 \right\} &= \frac{5}{9}(B_1 - B_0) : (V - \Gamma) \\ &\quad + \frac{5}{9}\mu^T \left[\frac{9}{10} B_1 (B_1 + \frac{9}{10} \text{I})^{-1} - B_0 \right] \mu. \end{aligned}} \quad (4.6)$$

Neither term in (4.6) has a fixed sign. This formula identifies the exact mean and covariance debts; it does not bound them.

Because (4.2)–(4.6) use only the complete endpoint, they survive internal representation-preserving subdivisions in the fixed common-heat scope. They do not license conditioning on an anticipative same-root random heat, nor do they combine distinct temporal roots.

5 The complete-cluster signed normal form

Let $\mathcal{C}$ be a contraction-closed output cluster, let $b_{\mathcal{C}}$ be its past-measurable baseline, let $Y_{\mathcal{C}}$ be the fully assembled current including all visits and future-feedback terms, and let $\tau_{\mathcal{C}}$ be its once-owned trace/forest scalar. Define

$$\mu_{\mathcal{C}} = \mathbb{E}[Y_{\mathcal{C}} \mid \mathcal{F}_{\text{past}}], \quad \Sigma_{\mathcal{C}} = \text{Cov}(Y_{\mathcal{C}} \mid \mathcal{F}_{\text{past}}). \quad (5.1)$$

Theorem 5.1 (exact signed cluster identity)

One has

$$\begin{aligned} \mathbb{E} \left[\frac{1}{2} \|b_{\mathcal{C}} + Y_{\mathcal{C}}\|^2 - \frac{1}{2} \tau_{\mathcal{C}} \mid \mathcal{F}_{\text{past}} \right] \\ = \frac{1}{2} \|b_{\mathcal{C}} + \mu_{\mathcal{C}}\|^2 + \frac{1}{2} \text{Tr} \Sigma_{\mathcal{C}} - \frac{1}{2} \mathbb{E}[\tau_{\mathcal{C}} \mid \mathcal{F}_{\text{past}}]. \end{aligned} \quad (5.2)$$

If $T_{\mathcal{C}}$ denotes the complete conditional trace operator with $\text{Tr} T_{\mathcal{C}} = \mathbb{E}[\tau_{\mathcal{C}} \mid \mathcal{F}_{\text{past}}]$, then the signed part is

$$\frac{1}{2} \text{Tr}(\Sigma_{\mathcal{C}} - T_{\mathcal{C}}). \quad (5.3)$$

The positive mean square and the signed covariance defect are coupled; a negative part may not be taken before they are added.

For the R-107 frozen/nonfrozen split, write

$$Y = b + A\xi + R(\xi), \quad S = AA^*, \quad (5.4)$$

and

$$L_{\text{full}} = \frac{1}{2} \|Y\|^2 - \frac{1}{2} \text{Tr} S - \frac{1}{2} \delta\tau. \quad (5.5)$$

Then (5.2) gives exactly

$$\mathbb{E}[L_{\text{full}} \mid \mathcal{F}_{\text{past}}] = \frac{1}{2} \|\mathbb{E}[Y \mid \mathcal{F}_{\text{past}}]\|^2 + \frac{1}{2} \text{Tr}(\text{Cov}(Y \mid \mathcal{F}_{\text{past}}) - S) - \frac{1}{2} \mathbb{E}[\delta\tau \mid \mathcal{F}_{\text{past}}]. \quad (5.6)$$

This is the smallest exact signed normal form obtained by the present audit. It is an identity, not positivity, a lower bound, or an exponential estimate.

6 Square the realized cluster covariance before averaging

The R-107 one-pair cluster is an exact order-of-operations test. Let

$$R = |z|^2 = \frac{\sigma^2}{2} T, \quad T \sim \text{Exp}(1), \quad P = R^2 - \sigma^2 R = \frac{\sigma^4}{4} (T^2 - 2T). \quad (6.1)$$

Since $\mathbb{E}T^n = n!$,

$$\mathbb{E}P = 0, \quad \mathbb{E}P^2 = \frac{\sigma^8}{2}, \quad \mathbb{E}P^3 = \frac{15\sigma^{12}}{4}. \quad (6.2)$$

The fourth cumulant is $213\sigma^{16}/4$, and therefore

$$\log \mathbb{E}e^{-qP} = \frac{q^2\sigma^8}{4} - \frac{5q^3\sigma^{12}}{8} + \frac{71q^4\sigma^{16}}{32} + O(\sigma^{20}). \quad (6.3)$$

Proposition 6.1 (average-before-square misses leading order)

For the realized independent-copy output covariance S_z ,

$$\mathbb{E}\|S_z\|_{\mathfrak{S}_2}^2 = \frac{5}{4}\sigma^8. \quad (6.4)$$

In the real output basis $(1, \sqrt{2}\cos 2x, \sqrt{2}\sin 2x)$,

$$\mathbb{E}S_z = \text{diag} \left(\frac{\sigma^4}{2}, \frac{\sigma^4}{4}, \frac{\sigma^4}{4} \right),$$

and hence

$$\|\mathbb{E}S_z\|_{\mathfrak{S}_2}^2 = \frac{3}{8}\sigma^8. \quad (6.5)$$

Consequently the three leading coefficients satisfy

$$\frac{q^2}{4} \frac{3\sigma^8}{8} = \frac{3q^2\sigma^8}{32} < \frac{q^2\sigma^8}{4} < \frac{5q^2\sigma^8}{16} = \frac{q^2}{4} \mathbb{E}\|S_z\|_{\mathfrak{S}_2}^2. \quad (6.6)$$

The averaged covariance misses $5q^2\sigma^8/32$ at leading order. The realized square has $q^2\sigma^8/16$ of leading room. The latter observation only preserves the route; it is not a general theorem.

For a finite check, take $q = 1$, $\sigma = 1/2$, and $A = q\sigma^4/4 = 1/64$. Direct integration gives

$$\mathbb{E}e^{-P} = 4\sqrt{\pi}e^{961/64}\operatorname{erfc}(31/8) = 1.0008509471256655\dots, \quad (6.7)$$

$$\log \mathbb{E}e^{-P} = 0.0008505852754225884\dots \quad (6.8)$$

The average-first cost is $3/8192 = 0.0003662109375$, while the square-first cost is $5/4096 = 0.001220703125$.

This does not by itself refute a determinant route with a sextic or nonlinear remainder. If the parent sextic is normalized as $\mathcal{Y} = 20R^3$, then $\mathbb{E}\mathcal{Y} = 15\sigma^6$ and

$$\log \mathbb{E}e^{-q(P+\zeta\mathcal{Y})} = -15q\zeta\sigma^6 + \frac{q^2\sigma^8}{4} + O(\zeta\sigma^{10} + \zeta^2\sigma^{12} + \sigma^{12}). \quad (6.9)$$

An average-first ledger must therefore satisfy

$$r + 15q\zeta\sigma^6 \geq \frac{5q^2\sigma^8}{32} + o(\sigma^8). \quad (6.10)$$

With $r = 0$ this forces

$$\zeta \geq \frac{q\sigma^2}{96} + o(\sigma^2), \quad (6.11)$$

which is $5\sigma^2/432 + o(\sigma^2)$ at $q = 10/9$.

7 Absolute future-feedback HS/PSD ledgers fail

Work on the normalized circle and let

$$G \sim N(0, 1), \quad h_M(G) = a \sin(MG), \quad f_N(x) = \cos(Nx), \quad z = h_M f_N. \quad (7.1)$$

For the scalar chart $\Phi(z) = z$, the nonlinear current is $C(z) = zD_x z$ and

$$\partial_G C = 2h_M h'_M f_N D_x f_N. \quad (7.2)$$

The exact spatial constants are

$$\|f_N\|_{H^2}^2 = \frac{(1+N^2)^2}{2}, \quad \|f_N\|_6^6 = \frac{5}{16}, \quad \|f_N D_x f_N\|_2^2 = \frac{N^2}{8}. \quad (7.3)$$

The last factor is supported at outputs $\{+2N, -2N\}$.

Proposition 7.1 (bounded budgets, divergent positive tangent)

Exact Gaussian trigonometric moments give

$$\mathbb{E}\|z\|_{H^2}^2 = \frac{a^2(1 - e^{-2M^2})(1 + N^2)^2}{4}, \quad (7.4)$$

$$\mathbb{E}\|z\|_6^6 = \frac{5a^6}{512}(10 - 15e^{-2M^2} + 6e^{-8M^2} - e^{-18M^2}), \quad (7.5)$$

but

$$\boxed{\mathbb{E}\|\Pi_{\{\pm 2N\}}\partial_G C\|_2^2 = \frac{a^4 M^2 N^2}{16}(1 - e^{-8M^2}).} \quad (7.6)$$

Thus the source energy and terminal sextic remain bounded in M , whereas the positive tangent square grows as M^2 . A rank-one HS square built from this tangent grows as M^4 ; before the fixed spatial factor its Gaussian coefficient is

$$\frac{a^8 M^4}{8}(3 - 4e^{-8M^2} + e^{-32M^2}). \quad (7.7)$$

The signed forest behaves differently:

$$\mathbb{E}(h'_M)^2 = \frac{a^2 M^2}{2}(1 + e^{-2M^2}), \quad (7.8)$$

$$\mathbb{E}(h_M h''_M) = -\frac{a^2 M^2}{2}(1 - e^{-2M^2}), \quad (7.9)$$

$$\mathbb{E}\{(h'_M)^2 + h_M h''_M\} = a^2 M^2 e^{-2M^2} \rightarrow 0. \quad (7.10)$$

Hence (7.6) rejects an absolute positive HS/PSD-Carleson ledger for arbitrary future-feedback selectors paid only by source energy and one terminal sextic. It does not reject the complete signed R-063 forest or the direct source action.

At $a = 1$, $N = 32$, $M = 1024$, the exact-underflow values are

$$X = 262656.25, \quad Y = 0.09765625, \quad \mathbb{E}[X(G)^{1/2}Y(G)^{1/2}] = 151.93755945340263\dots, \quad (7.11)$$

while the tangent square is 67108864. The two separated second-jet terms are 524288 and -524288 to machine precision.

The original R-088 strict-past theorem has $D_j a_k = 0$ and is not refuted by this fixture. The failure concerns an attempted same-root future-feedback extension after R-107.

8 A nonvacuous statement of the remaining theorem

The direct formulation is already the cleanest legal target. There must exist fixed constants

$$0 \leq \eta < \frac{9}{20}, \quad 0 \leq \zeta < \frac{3}{20}, \quad C < \infty, \quad (8.1)$$

independent of cutoff, temporally faithful chart, admissible control, and representation-preserving subdivision, such that

$$\boxed{\sum_k \mathbb{E} P_k^{\text{comp}} \geq -\eta \mathbb{E} \sum_k \|h_k\|^2 - \zeta \mathbb{E} \|Z_K\|_6^6 - C.} \quad (8.2)$$

Here P_k^{comp} contains complete heat, covariance trace, mixed baseline, future feedback, rational recovery, all visits, and the full R-063 forest exactly once. Equation (8.2) is **TARGET, NOT THEOREM**.

A determinant-compatible route must first form a legal contraction-closed conditional cluster and then prove a genuine one-step normalizer inequality of the form

$$\begin{aligned} \log \mathbb{E} \left[e^{-q(\mathfrak{P}_{j,C} + \eta_{j,C} \mathcal{E}_j + \zeta_{j,C} \mathcal{Y}_{\text{parent}(j)})} \mid \mathcal{F}_{j-1} \right] \\ \leq \frac{q^2}{4} \|S_{j,C}^{\text{real}}\|_{\mathfrak{S}_2}^2 + r_{j,C}. \end{aligned} \quad (8.3)$$

The data on the right must be legal at the conditioning step, $r_{j,C} \geq 0$, and not chosen after observing the inequality's deficit. One unambiguously sufficient, though possibly stronger than necessary, global specification is a pathwise or deterministic ledger

$$\sum_{j,C} \left\{ \frac{q^2}{4} \|S_{j,C}^{\text{real}}\|_{\mathfrak{S}_2}^2 + r_{j,C} \right\} \leq C + \epsilon \sum_j \|h_j\|^2 + \delta \|Z_K\|_6^6, \quad (8.4)$$

uniformly in the same four parameters, with

$$q \sum_{j,C} \eta_{j,C} + \epsilon < \frac{1}{2}, \quad q \sum_{j,C} \zeta_{j,C} + \delta < \frac{1}{6}. \quad (8.5)$$

An equivalent conditional-supermartingale ledger is also admissible. A mere finite-cutoff assertion $\sum r < \infty$ is vacuous because r could be defined to equal the observed deficit. Equations (8.3)–(8.5) are **TARGETS, NOT THEOREMS**.

Proposition 6.1 fixes the order in (8.3): form the realized legal conditional cluster covariance, square its HS norm, and only then take the outer expectation. Replacing it by $\|\mathbb{E}S\|^2$ is false without the explicit remainder/sextic tradeoff (6.10).

Exact $q = 10/9$ budget audit

The conservative R-103 regular allocation supplies the following arithmetic witness (8.6), not a progressive theorem:

	action scale	after multiplication by $q = 10/9$
full source coefficient	9/20	1/2
full sextic coefficient	3/20	1/6
η_*	1/440	1/396
ζ_*	3/100	1/30
source reserve	197/440	197/396
sextic reserve	3/25	2/15
each of seven source shares	1/3080	1/2772
each of seven sextic shares	3/700	1/210
each of eight source shares	1/3520	1/3168
each of eight sextic shares	3/800	1/240.

In particular

$$\frac{1}{2} - \frac{1}{396} = \frac{197}{396}, \quad \frac{1}{6} - \frac{1}{30} = \frac{2}{15}. \quad (8.7)$$

These positive reserves show that q -inflation itself is not the current obstruction. No assertion is made that the progressive theorem must fit the particular trial values η_*, ζ_* .

9 Proof-route ledger

1. Historical R-085 weighted Cartan bridge: superseded as the exact current target and still unproved. R-087's regular spatial half remains valid.

2. Direct R-088 unweighted sequential bridge: open. It is not refuted by the future-feedback fixture because its strict-past derivative hypothesis is deliberately violated.
3. Historical $F_{6.5}$ or fixed-chart K_R temporalized visit by visit: failed by the exact sign-changing subdivision table (3.11).
4. Complete endpoint conditional mean/covariance: advanced as the exact quotient-safe identities (4.2) and (4.4). Their sign is open.
5. Complete-cluster signed normal form: advanced as (5.2)–(5.6). It prevents premature separation but supplies no lower bound.
6. Average covariance before HS square: failed at both leading order and the finite sample (6.7)–(6.8), unless an explicit nonlinear/sextic remainder obeys (6.10).
7. Realized cluster square before outer averaging: survives the one-pair test with room. This is viability evidence, not a theorem.
8. Absolute future-feedback HS/PSD ledger from source energy plus one sextic: failed by (7.4)–(7.10). Signed forest cancellation survives.
9. Direct uniform complete source action (8.2) and the legally normalized determinant route (8.3)–(8.5): open.

These entries record reusable route decisions, not private search traces.

10 Devil’s-advocate audit

1. **Objection: the direct R-088 constant is small enough to close the progressive theorem.**
UPHELD AS UNSUPPORTED. The smaller constant changes the sufficient spatial summation but does not prove the expectation-inside sequential bridge or control the same-root future-feedback selector.
2. **Objection: $F_{6.5}$ may still be estimated on every visit and summed conservatively.**
UPHELD AGAINST THE CLAIM. It changes from $-992/81$ to $427/162$ on two representations of the same endpoint. The complete endpoint, not $F_{6.5}$, is the quotient coordinate.
3. **Objection: retaining both $\mathcal{R}_Q$ and $F_{6.5}$ gives extra safety.**
UPHELD AGAINST THE CLAIM. Equation (3.7) shows that $L : Q/2$ is already inside $\mathcal{R}_Q$; the proposed safety is double counting.
4. **Objection: (4.6) proves positivity after optimizing the control.**
UPHELD AGAINST THE CLAIM. Both the mean bracket and the covariance pairing are sign-indefinite. Exact scalar fixtures attain both signs.
5. **Objection: the cluster covariance can be averaged before applying the determinant because Jensen is conservative.**
UPHELD AGAINST THE CLAIM. Squaring is convex in the wrong order for this purpose. The averaged quantity supplies $3/32$ while the exact log-Laplace term needs $1/4$ at leading order.
6. **Objection: the one-pair failure disproves every determinant route.**
DISMISSED. The realized square supplies $5/16$ and therefore has leading room. A legal sequential conditional determinant remains possible.

7. **Objection: the oscillatory tangent disproves the signed source action.**

DISMISSED. The separated M^2 second jets cancel to $a^2 M^2 e^{-2M^2}$. Only the absolute positive-tangent architecture is refuted.

8. **Objection: summability at each finite cutoff makes (8.3) nonvacuous.**

UPHELD AGAINST THE CLAIM. A post-hoc remainder can always equal the finite deficit. Uniform dependence, sign, measurability, and a cutoff/chart/ control/subdivision-independent ledger are essential.

11 Authorities and scope firewall

The load-bearing authorities are R-063 (complete Taylor/Wick forest), R-085 (historical weighted Cartan and rational forms), R-087 (regular spatial half and failed translated-model extraction), R-088 (exact direct unweighted OU target), R-103 (regular H_N /REG allocation), R-104 (fixed-chart endpoint assembly), R-105 v1.1 (subdivision quotient), and R-107 (coherent cluster and direct source-action frontier).

Proved here are only the authority audit, the exact subdivision table, the complete endpoint conditional and optimized identities, the abstract complete-cluster mean/covariance identity, the one-pair square-before-average obstruction and quantified sextic tradeoff, the arbitrary-selector absolute future-feedback obstruction, and the exact arithmetic budget.

Not proved here are the direct bridge (2.5)–(2.6), any sign for (4.6), the uniform lower bound (8.2), the determinant targets (8.3)–(8.5), an arbitrary same-root random-heat extension, progressive/revisit H_A , OVERLAP_{src}, Nelson, cutoff or floor removal, an interacting measure, Sector-A closure, or a T5–T7 promotion.

12 Executable verification

The primary certificate derives all rational owners and conditional forms in exact symbolic algebra and derives every Gaussian moment from its generating identities. The independent certificate imports no symbolic, array, scientific, or primary module. It uses a second-order rational dual number, exact finite distributions, factorial cumulants, hand matrix algebra, the closed erfc moment, and direct Gaussian quadrature.

Run

```
E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_complete_cluster_quotient_carleson_frontier_verify.py
```

from the repository root. The integrated verifier reruns both certificates and checks the note, PDF, manifest, authority pins, negative and exploration ledgers, public surfaces, and the no-overclaim firewall.

Result footer

```
Result ID: A13-CLASSII-COMplete-CLUSTER-QUOTIENT-CARLESON-FRONTIER
Claim:     A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION
Ledger ID: R-108
Status:    T4 ANALYTIC/EXACT/EXECUTED SYNTHESIS AND BOUNDARY;
           NELSON AND SECTOR A OPEN
Scope:     Finite cutoff, fixed positive floor, q=10/9; exact historical-
           atom quotient audit, complete endpoint/cluster identities, and
           scoped order-of-operations method fixtures.
Precise statement: historical R-085 (4.11) is not the exact current target;
           F_6.5 and visitwise K_R do not descend under subdivision; the
           complete endpoint has exact conditional mean/covariance and CM-
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optimized forms; realized covariance must be squared before the
 outer average; absolute arbitrary future-feedback HS ledgers fail.
 Dependencies: R-063, R-085, R-087, R-088, R-103, R-104, R-105, R-107.
 Evidence grade: ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.
 Reproduction command:
 E:\Dev\TECT.venv\Scripts\python.exe '
 codes/foundations/a13_classii_complete_cluster_quotient_carleson_frontier_verify.py
 Expected output:
 primary 66/66; independent 90/90; integrated 230/230;
 aggregate 386/386 PASS.
 Negative results: NG-2026-07-28-A13-RATIONAL-TAYLOR-OWNER-SUBDIVISION;
 NG-2026-07-28-A13-AVERAGED-COVARIANCE-BEFORE-HS-SQUARE;
 NG-2026-07-28-A13-ABSOLUTE-FUTURE-FEEDBACK-CARTAN-CARLESON;
 NG-2026-07-28-A13-ADAPTED-SECOND-JET-TERMSEPARATION.
 Falsification gate: failure of any quotient, conditional endpoint, cluster,
 moment, covariance-order, feedback, budget, evidence, hash, PDF,
 ledger, or scope assertion.
 Tier before / after: T4 / T4; no promotion.
 No-overclaim statement:
 identities are re-expressions, not lower bounds; the complete
 adapted cluster/source-action inequality, OVERLAP_src, Nelson,
 removals, measure, and Sector A remain open.
 Next required action: prove the cutoff/chart/control/subdivision-uniform
 signed complete-cluster lower bound (8.2), or a legally normalized
 square-before-average conditional determinant implying it.